BCSE429P

Cyber Physical Systems Design Lab

Laboratory 5 - Three state FSM model for a Thermostat

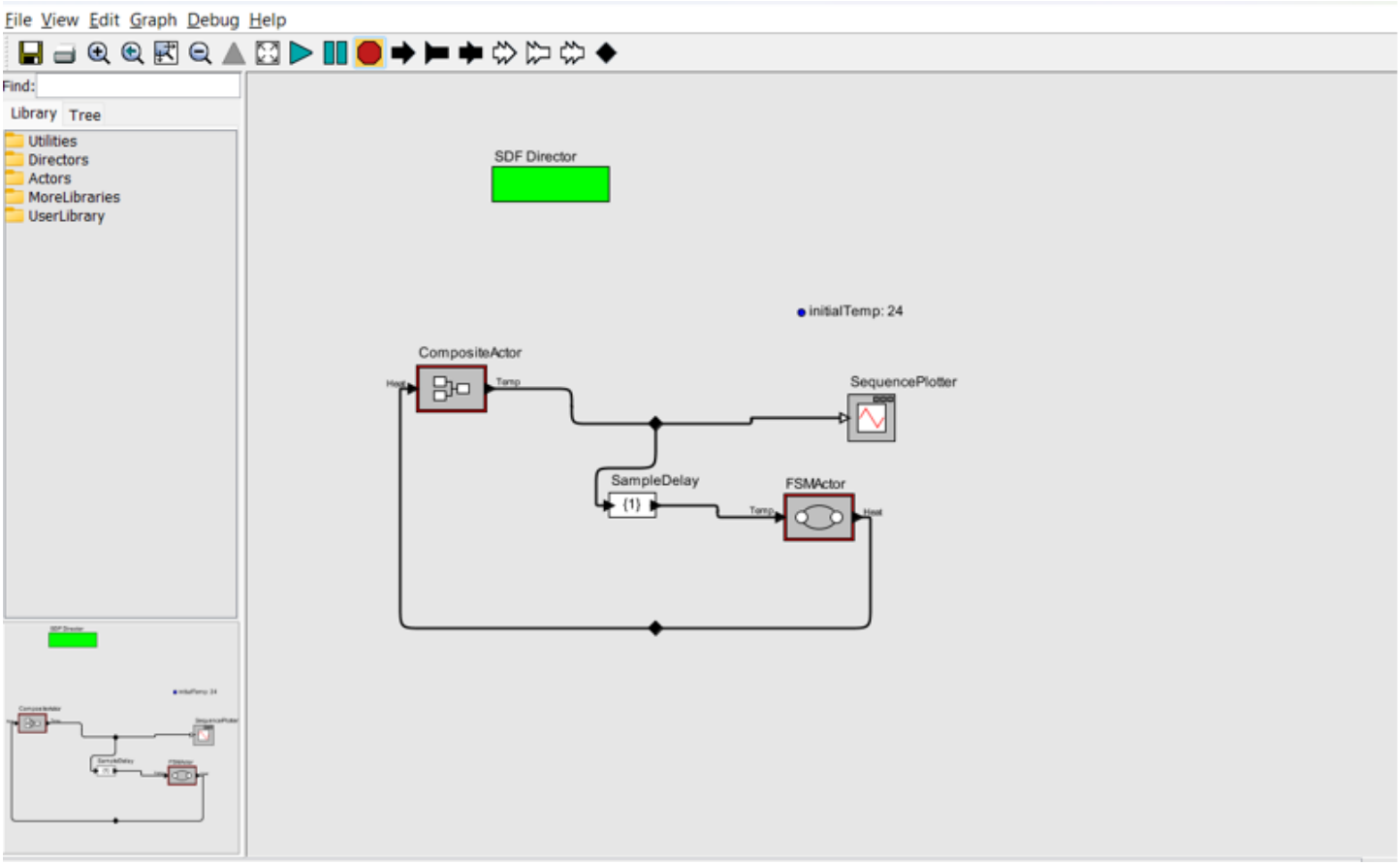
Name: Aman Kumar Singh

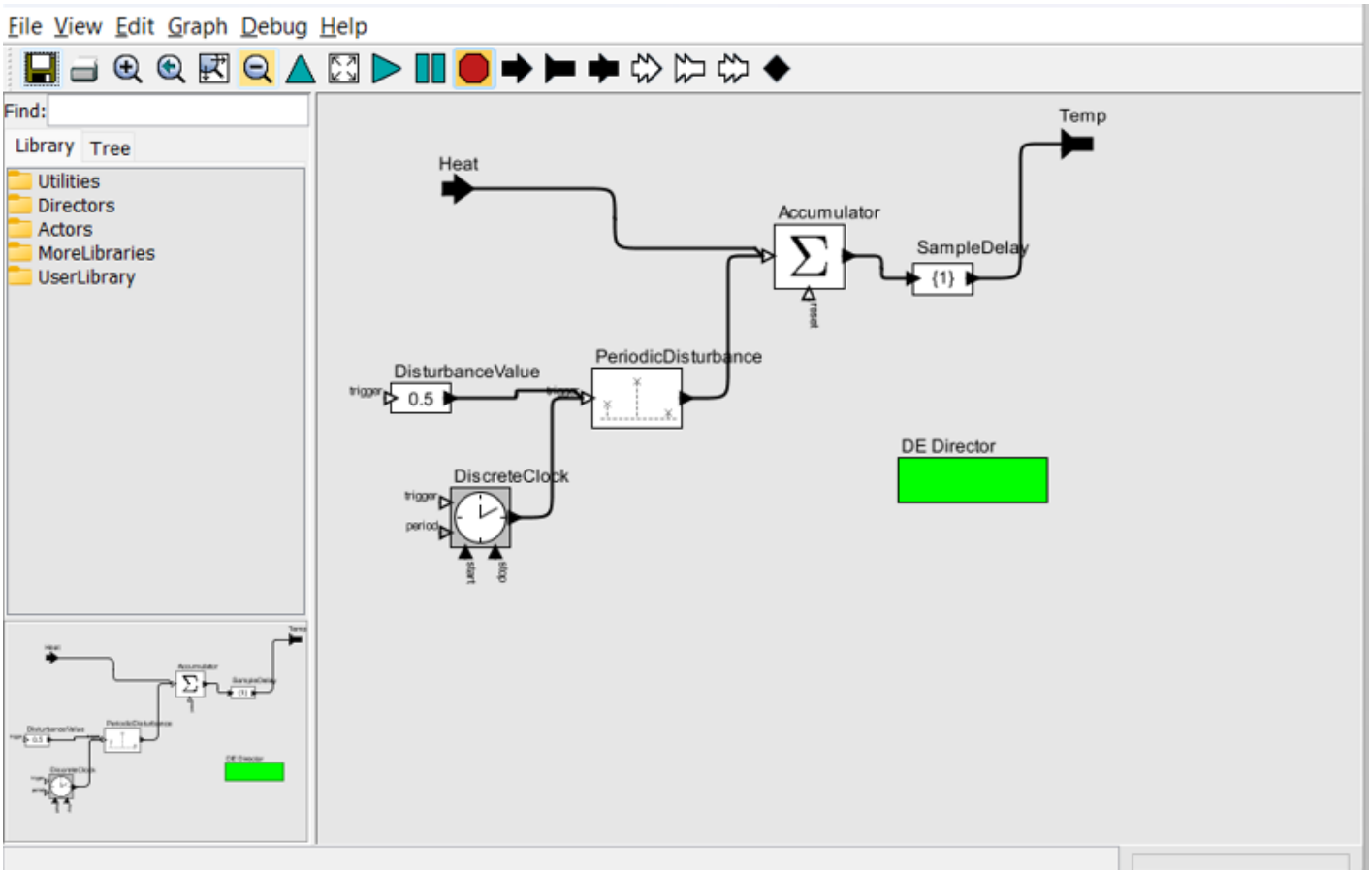
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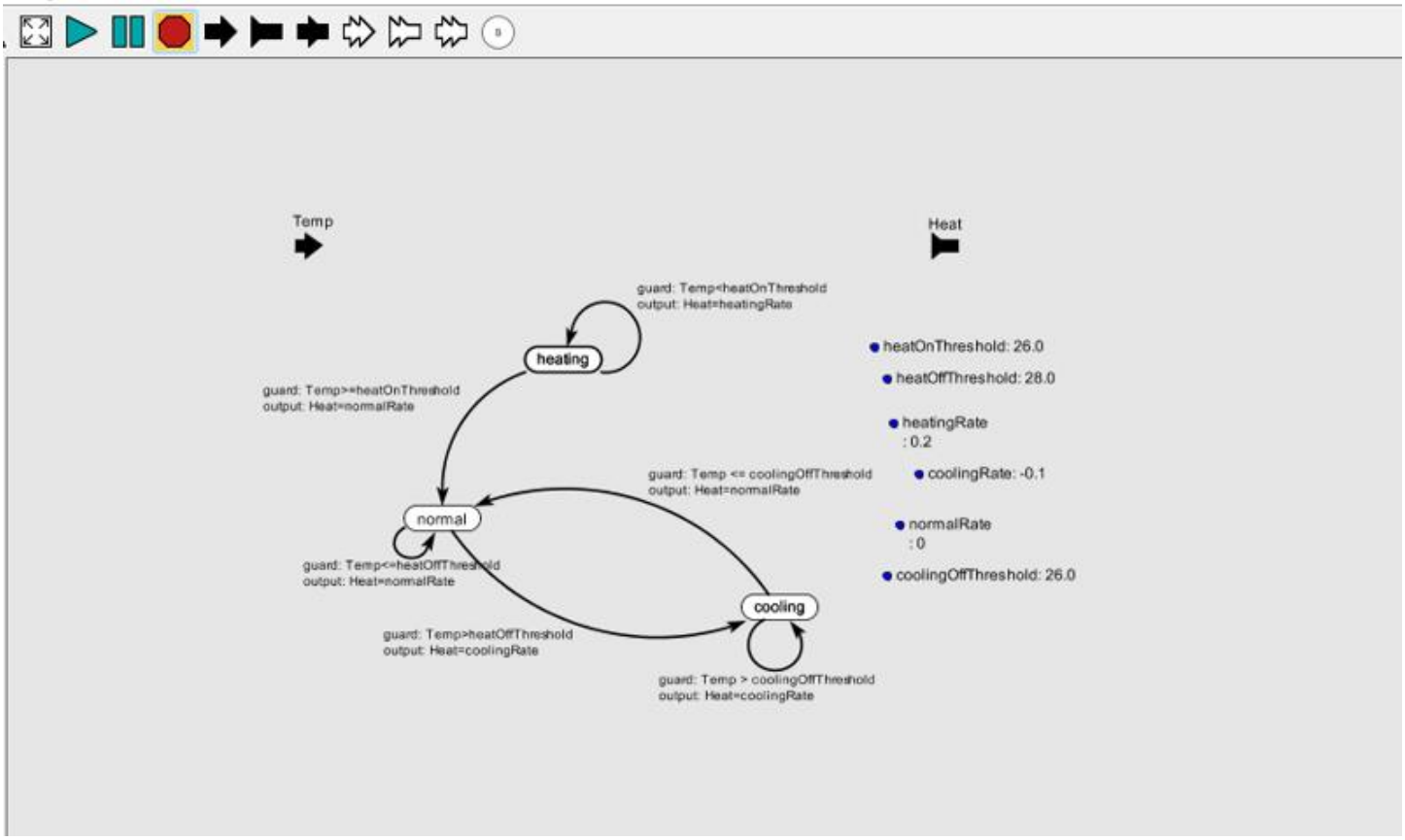
Question:

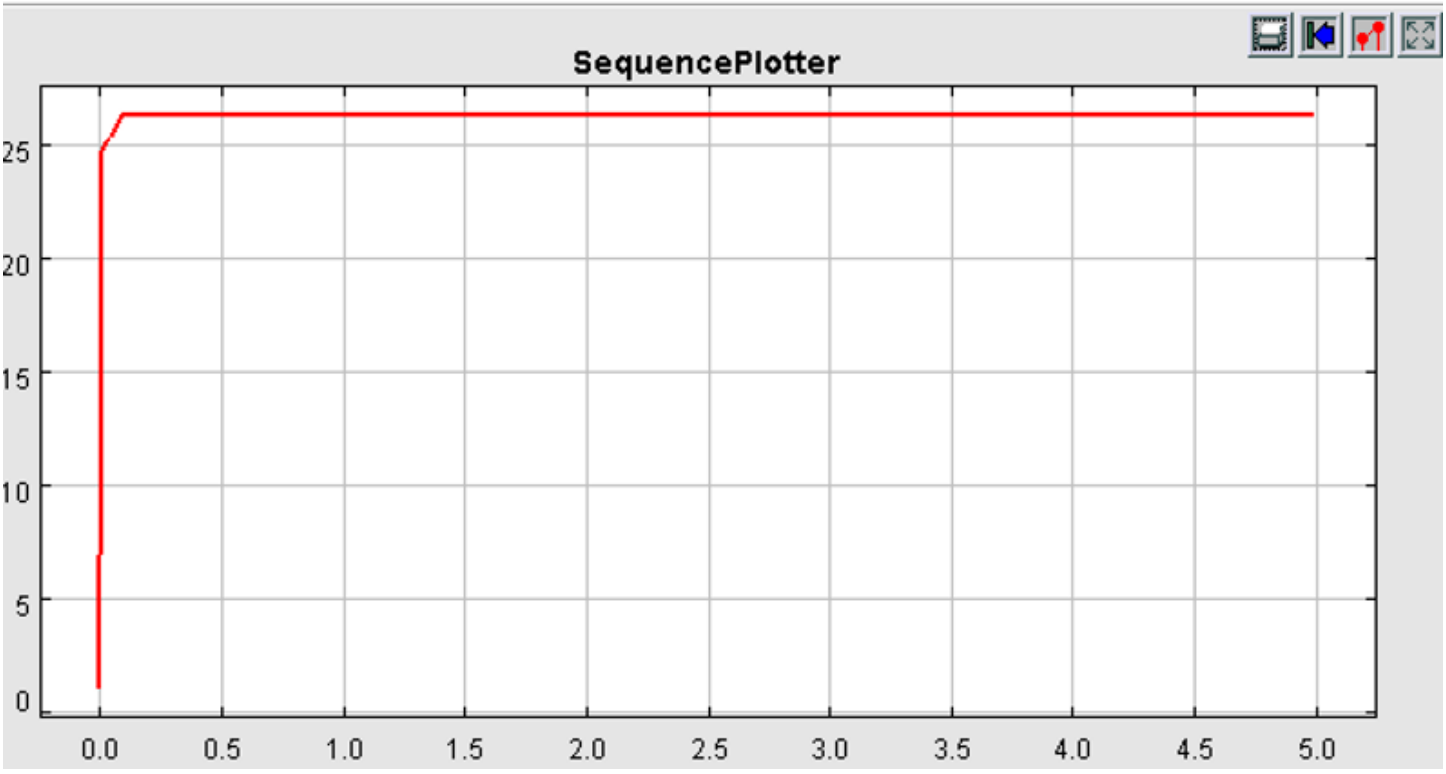
Design and implement a three-state Finite State Machine (FSM) model of a thermostat using Ptolemy II.

1. The thermostat shall have three states: Heating, Normal, and Cooling.
2. The initial temperature is 24°C, and the initial state of the thermostat is Heating.
3. When the thermostat is in the Heating state, the temperature shall increase by 0.2°C per sample.
4. When the temperature reaches 26°C, the thermostat shall transition from Heating to Normal.
5. In the Normal state, the temperature shall remain unchanged at 26°C.
6. To model environmental variation, introduce an external temperature disturbance that increases the temperature by 0.5°C at regular intervals. When the temperature becomes greater than 28°C, the thermostat shall transition to the Cooling state.
7. When the thermostat is in the Cooling state, the temperature shall decrease by 0.1°C per sample.
8. When the temperature falls to 26°C, the thermostat shall transition from Cooling back to Normal.
9. The thermostat should therefore demonstrate the following state transitions:
   Heating → Normal → Cooling → Normal
10. Implement the FSM using FSMActor and appropriate Ptolemy II actors such as Composite Actor, AddSubtract, Constant, Clock/Periodic signal, and Sequence Plotter, wherever required.
11. Simulate the model for a total of 500 samples.
12. Use a Sequence Plotter to display the temperature variation with respect to the sample number.
13. Clearly observe and verify the state transitions and temperature changes during the simulation.


File View Edit Graph Debug Help
Find:
Library Tree
Utilities
Directors
Actors
MoreLibraries
UserLibrary
SDF Director
initialTemp: 24
CompositeActor
Heat
Temp
SequencePlotter
SampleDelay
{1}
FSMActor


File View Edit Graph Debug Help
Find:
Library Tree
Utilities
Directors
Actors
MoreLibraries
UserLibrary
Heat
Accumulator
Temp
SampleDelay
{1}
reset
DisturbanceValue
trigger
0.5
PeriodicDisturbance
DE Director
DiscreteClock
trigger
period
start
stop


Temp
Heat
guard: Temp<heatOnThreshold
output: Heat=heatingRate
heating
guard: Temp>=heatOnThreshold
output: Heat=normalRate
heatOnThreshold: 26.0
heatOffThreshold: 28.0
heatingRate
: 0.2
guard: Temp <= coolingOffThreshold
output: Heat=normalRate
coolingRate: -0.1
normal
normalRate
: 0
guard: Temp<=heatOffThreshold
output: Heat=normalRate
coolingOffThreshold: 26.0
cooling
guard: Temp>heatOffThreshold
output: Heat=coolingRate
guard: Temp > coolingOffThreshold
output: Heat=coolingRate


SequencePlotter
25
20
15
10
5
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0